



## COS 132 – Imperative Programming

# Foundation Test 1

ST2 Programme

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| <b>Module:</b>      | COS 132 – Imperative Programming   |
| <b>Assessment:</b>  | Foundations Test 1 (ST2 Programme) |
| <b>Set by:</b>      | Swift Tuition                      |
| <b>Duration:</b>    | 30 minutes                         |
| <b>Total Marks:</b> | 22                                 |
| <b>Difficulty:</b>  | Easy – diagnostic baseline         |

**Programme note.** This material is for Day 1 of the 7-day Swift Tuition COS 132 programme. The premium course is available at [www.swifttuition.com/tuks/cos132](https://www.swifttuition.com/tuks/cos132). Learners watching before ST2 2026 may use coupon code **200COS132** at checkout for R200 off the full price. All three foundation tests are included; one test can be taken now, and the remaining tests can be taken anytime after ST2. Where units are referenced, they are Swift Tuition learning units based on how the module's lecture content was split, covering lectures 1–16.

## 1 Instructions

1. Read each question carefully and answer all of them.
2. Write your answers in a separate file or on paper. Do not edit this question paper.
3. The assessment comprises **4** questions over **22** marks.
4. This is a closed-book assessment. No internet, no notes, no compiler.
5. You have **30 minutes** to complete the paper.

|           |   |   |   |   |       |
|-----------|---|---|---|---|-------|
| Question: | 1 | 2 | 3 | 4 | Total |
| Marks:    | 8 | 6 | 4 | 4 | 22    |

Name and surname: \_\_\_\_\_

Student number: \_\_\_\_\_

## 2 Question 1 [8 marks]

(a)

For each of the statements below, state whether the statement is **True** or **False**.

- i. The `cin` operator can read a string containing spaces directly into a `string` variable. (1)
- ii. The `<iomanip>` header must be included to use the `setw` function. (1)
- iii. In C++, the operator `=` is used for assignment, and `==` is used for comparison. (1)
- iv. A function in C++ can have a default value for its parameters, but only for the leftmost parameters. (1)

(b)

For each of the questions below, choose the most correct option.

- i. Which of the following C++ statements correctly declares a constant for the value of pi? (1)
  - A. `int PI = 3.14159;`
  - B. `const double PI = 3.14159;`
  - C. `double const PI;`
  - D. `define PI 3.14159;`
- ii. Which one of the following expressions evaluates to 7? (1)
  - A. `15 / 2`
  - B. `15 % 2`
  - C. `15.0 / 2`
  - D. `15 / 2.0`
- iii. Which header file must be included to use `cout` and `cin`? (1)
  - A. `<stdio.h>`
  - B. `<iostream>`
  - C. `<string>`
  - D. `<iomanip>`
- iv. Which of the following is the correct way to declare a function that takes two integers and returns their sum? (1)
  - A. `int add(int, int);`
  - B. `void add(int a, int b);`
  - C. `add(int a, int b): int;`
  - D. `function int add(int a, int b);`

### 3 Question 2 [6 marks]

For each of the following programs, study the code and provide the exact output the program would produce.

(a)

```
#include <iostream>
using namespace std;

int main() {
    int a = 7;
    int b = 3;
    cout << a / b << " " << a % b;
    return 0;
}
```

(2)

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(b)

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;
    int y = x++;
    int z = ++x;
    cout << x << " " << y << " " << z;
    return 0;
}
```

(2)

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(c)

```
#include <iostream>
using namespace std;

int compute(int n) {
```

```
    return n * 2;
}

double compute(double n) {
    return n + 0.5;
}

int main() {
    cout << compute(4) << " " << compute(4.0);
    return 0;
}
```

(2)

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## 4 Question 3 [4 marks]

Consider the following Logo procedure:

```
TO PRINT_LINE :N
  REPEAT :N [
    PRINT "*"
  ]
END
```

The procedure prints `:N` lines, each containing a single asterisk.

Convert the Logo procedure into an equivalent C++ function called `printLine` that takes one `int` parameter and produces the same output. You do **not** need to write a `main` function. Assume `iostream` and the `std` namespace are available.

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## 5 Question 4 [4 marks]

Write **single C++ statements** that perform each of the following tasks. You may assume any necessary headers are included and that the `std` namespace is in use.

(a)

Declare an integer variable called `count` and initialise it to zero. (1)

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(b)

Read a full line of text (including spaces) from standard input into a `string` variable called `name`. (1)

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(c)

Print the value of a `double` variable called `total` with exactly **2** digits after the decimal point. (1)

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(d)

Calculate the average of two `int` variables `a` and `b` as a `double`, and store the result in a `double` variable called `avg`. (Use a type cast.) (1)

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*End of paper – 22 marks – 30 minutes*